

MetaboPrep Data Preparation Summary Report

27 March, 2025

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1 Project Information

This metaboprep2 report summarizes the data preparation steps for:

- **Project:** MYPROJECT
- **Platform:** metabolon_v1
- **Data processing date:** 2025-03-26

1.1 Overview

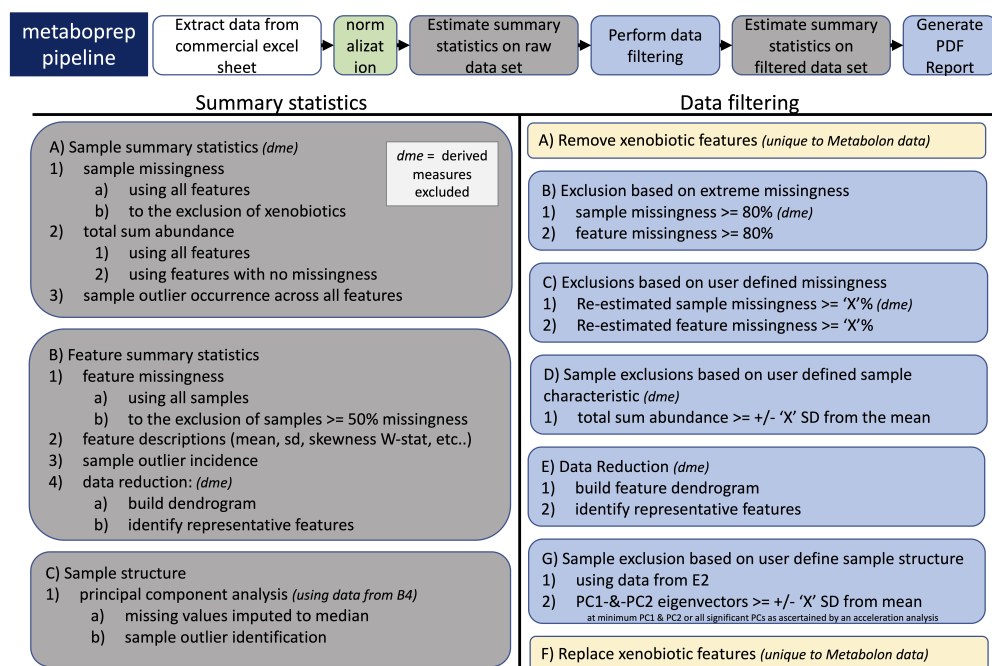
The metaboprep R package performs three key operations:

1. **Assessment & Summary Statistics:** Provides an initial assessment of raw metabolomics data.
2. **Data Filtering:** Applies filtering techniques to clean the dataset.
3. **Post-Filtering Assessment:** Evaluates the filtered dataset, particularly in relation to batch variables when available.

This report contains descriptive information on both raw and filtered metabolomics data for **MYPROJECT**.

Please raise any issues on the [GitHub issues](#) page.

1.2 Data preparation workflow:



2 Summary of raw data

2.1 Sample size of MYPROJECT data set:

Table 1: Sample Size Table

Dataset	Samples	Features
RAW	100	100
SCALED	100	100
POST QC	98	97

3 Missingness

Missingness is evaluated across samples and features using the original/raw data set.

3.1 Visual structure of missingness in the raw data

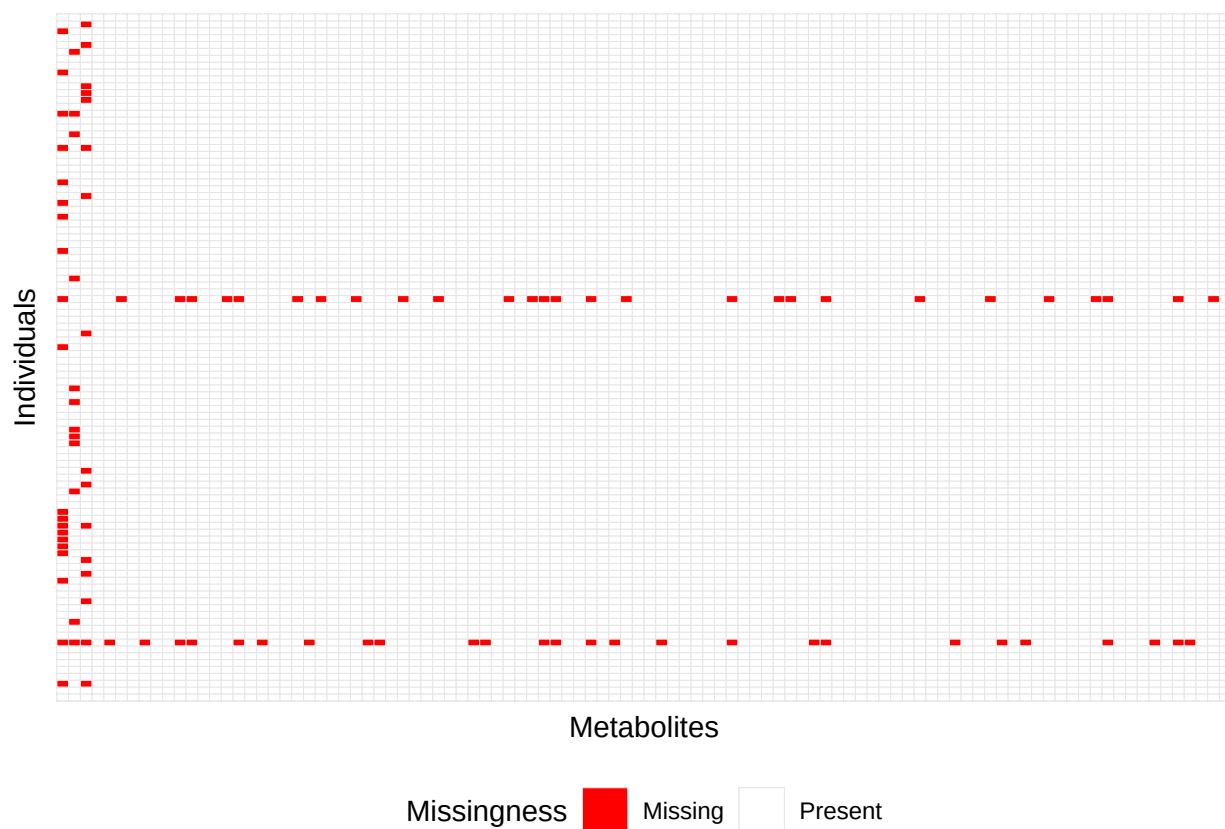


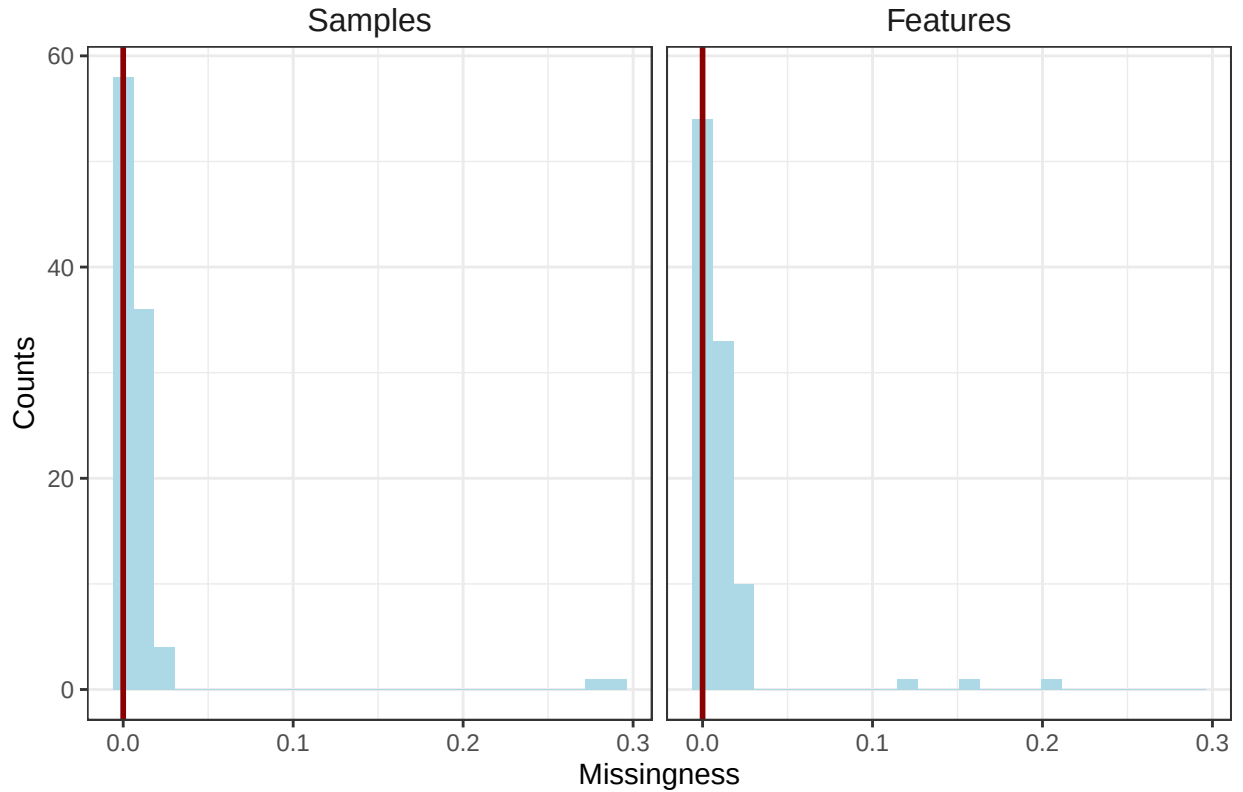
Table 2: Table of sample and feature missingness percentiles.

Percentile	Samples	Features
0.00	0.00	0.00
0.25	0.00	0.00
0.50	0.00	0.00
0.75	0.01	0.01
1.00	0.29	0.20

Table 3: Estimates of samples sizes under various levels of feature missingness.

Missingness (%)	Sample Size
10%	90
20%	80
30%	70
40%	60
50%	50

3.2 Summary of sample and feature missingness



Distribution of missingness with median illustrated by the red vertical line.

Table 4: Sample exclusion summary

Exclusion Reason	Count
derived feature	0
xenobiotic feature	98
extreme sample missingness	0
extreme feature missingness	0
user defined sample missingness	2
user defined feature missingness	0
user defined sample totalpeakarea	0
user defined sample pca outlier	0
outlier udist turned na	0
outlier udist winsorized	0

Note:

Six primary data filtering exclusion steps were made during the preparation of the data. User-defined thresholds indicated with an asterisk.

¹ Samples with missingness $\geq 80\%$.

² Features with missingness $\geq 80\%$ (xenobiotics are not included in this step).

³ Sample exclusions based on the user defined threshold of $\geq 20\%$.

⁴ Feature exclusions based on user defined threshold of $\geq 20\%$, (xenobiotics are not included in this step).

⁵ Samples with a total-peak-area or total-sum-abundance that is ≥ 5 standard deviations from the mean.

⁶ Samples that are ≥ 5 standard deviations from the mean on principal component axis 1 and 2

4 2. Data Filtering

4.1 Exclusion summary

4.2 Metabolite or feature reduction and principal components

5 3. Summary of filtered data

5.1 Sample size (N)

5.2 Relative to the raw data

5.3 Structure among samples

5.4 Feature Distributions

5.4.1 Estimates of normality: W-statistics for raw and log transformed data

5.5 Outliers

5.5.1 Notes on outlying samples at each metabolite|feature

6 4. Variation in filtered data by available variables

6.0.1 feature missingness

6.0.2 sample missingness

6.1 Multivariate evaluation: batch variables

7 5. Total peak or abundance area (TA) of samples:

7.1 Relationship with missingness

7.2 Univariate evaluation: batch effects

7.2.1 Multivariate evaluation: batch variables

8 6. Power analysis